

HDOC DAY-32

Title: Matrix Commutative Properties via Variable Length Arrays (VLAs) in C

1. Problem Statement

Write a C program to perform the following menu-driven operations using separate functions:

1. Get two variable length 2D arrays and prove that matrix addition is commutative ($A + B = B + A$).
2. Get two variable length 2D arrays and prove that matrix multiplication is not commutative ($A \times B \neq B \times A$).
3. Exit.

2. Theoretical Background & Mathematical Proof

1. Matrix Addition Commutative Property:

Matrix addition is performed element-wise. Since integer addition in real numbers is commutative ($a_{ij} + b_{ij} = b_{ij} + a_{ij}$), the resulting sum matrices $A + B$ and $B + A$ are identical in dimensions and corresponding values.

2. Matrix Multiplication Non-Commutative Property:

Matrix multiplication involves row-by-column dot products ($C_{ij} = \sum a_{ik} b_{kj}$). Even when dimensions allow both $A \times B$ and $B \times A$ to be computed, the calculated dot products differ due to the directional nature of matrix multiplication, making $A \times B \neq B \times A$ in general.

3. Test Case Table

Test Case	Option	Input Matrices	Computation / Comparison	Outcome	Status
1	Addition	$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix},$ $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$	$A + B = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$		

$$B + A = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix} \mid A + B == B + A$$

PROVED | Pass | | 2 | Multiplication | $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} \mid A \times B = \begin{bmatrix} 4 & 6 \\ 10 & 12 \end{bmatrix}$

$$B \times A = \begin{bmatrix} 2 & 4 \\ 10 & 14 \end{bmatrix} | A \times B \neq B \times A$$

PROVED | Pass | | **3** | Exit | Choice 3 | Exit system call |
 Program Terminated | Pass |

4. Source Code Listing

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PROVED | Pass | | 3 | Exit | Choice 3 | Exit system call | Program Terminated | Pass |

4. Source Code Listing

4. Execution Screenshots

```
kali@kali:~/HDOC_Day32$ gcc day32.c
kali@kali:~/HDOC_Day32$ ./day32
Session Actions Edit View Help
kali@kali:~/HDOC_Day32$
kali@kali:~/HDOC_Day32$ cd HDOC_Day32
kali@kali:~/HDOC_Day32$
kali@kali:~/HDOC_Day32$ nano day32.c
kali@kali:~/HDOC_Day32$
kali@kali:~/HDOC_Day32$ gcc day32.c -o day32
kali@kali:~/HDOC_Day32$ ./day32

===== MATRIX COMMUTATIVITY DEMO =====
1. Prove Matrix Addition is Commutative ( $A + B = B + A$ )
2. Prove Matrix Multiplication is NOT Commutative ( $A * B \neq B * A$ )
3. Exit
Enter your choice (1-3): 1

--- 1. Prove Matrix Addition is Commutative ( $A + B = B + A$ ) ---
Enter rows and columns for both matrices: 2 2
Enter elements of Matrix A (2x2):
A[0][0]: 1 2 3 4
A[0][1]: A[1][0]: A[1][1]: Enter elements of Matrix B (2x2):
B[0][0]: 5 6 7 8
B[0][1]: B[1][0]: B[1][1]:
Matrix A + B:
6      8
10     12

Matrix B + A:
6      8
10     12

PROVED: Matrix Addition is Commutative ( $A + B = B + A$ ).

===== MATRIX COMMUTATIVITY DEMO =====
2. Prove Matrix Addition is Commutative ( $A + B = B + A$ )
2. Prove Matrix Multiplication is NOT Commutative ( $A * B \neq B * A$ )
3. Exit
Enter your choice (1-3):
```